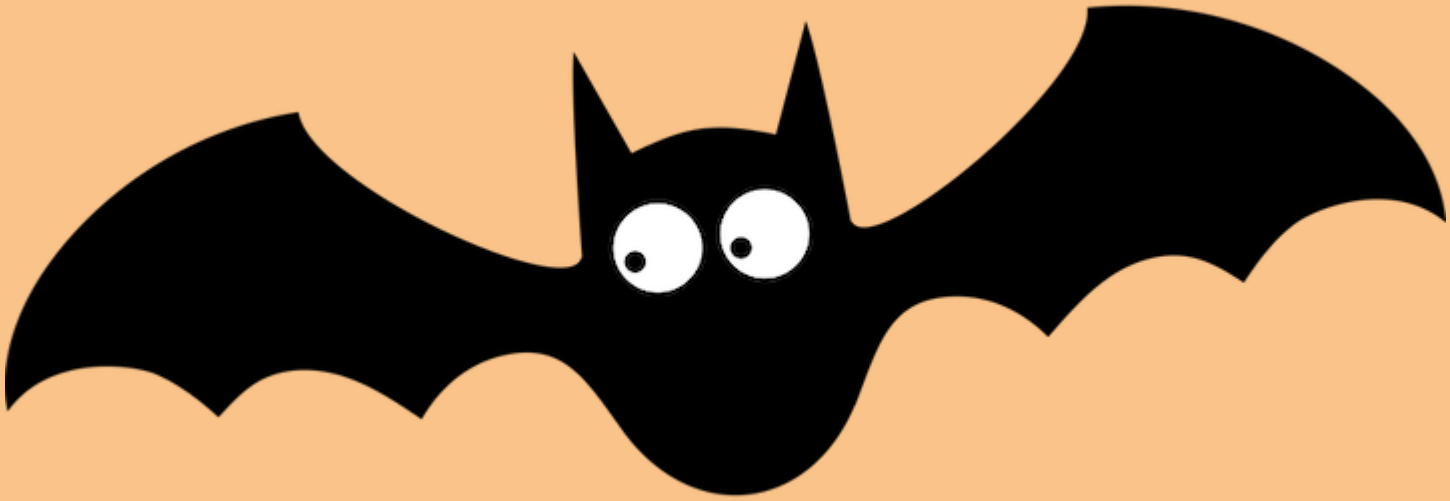


Gradient Boosting

Son Nguyen

Gradient Boosting

* Combine of multiple "weak" models (usually simple trees)

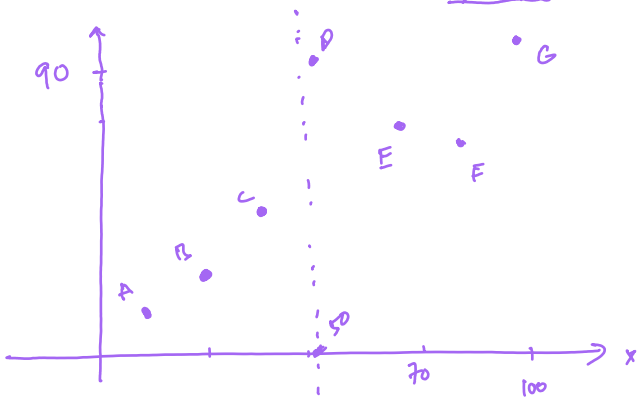
Example :

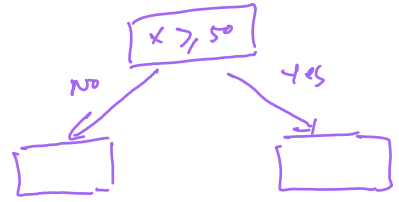
temp today → x, temp tomorrow → y

x	y
70	80
60	90
30	20
40	40
80	75
90	100
10	15

This is a regression (not classification) b/c y is continuous / numeric

* How decision trees work for regression





A, B, C
predict by the average y values of A B C

D, E, F, G
predict by the average values of D E F G

Model 1 is train on the original data

x	y	y_1	e_1
70	50	100	-20
60	90	20	70
30	20	30	-10
40	40	70	-30
80	75	100	-25
90	100	90	10
10	15	25	-10

y_1 : prediction of Model 1 (M1)

e_1 : errors of M1 : $y - y_1$

target of model 2

$= e_1 - \hat{e}_1$

x	e_1	$\hat{e}_1$	e_2
70	-20	-15	-5
60	70	80	-10
30	-10	-5	-5
40	-30	-30	0
80	-25	-20	-5
90	10	15	-5
10	-10	-10	0

Data of Model 2

Model 2 is designed to predict the error of M1

$\hat{e}_1$: prediction of M2

target of M3

x	e_2	$\hat{e}_2$	e_3
70	-5	-7	2
60	-10	-5	-5
30	-5	-6	1
40	0	2	-2
80	-5	-8	3
90	-5	-2	-3
10	0	-1	1

$\rightarrow \cdots \rightarrow \cdots$

Let say we stop at M3, the final prediction of the committee of M1, M2, M3 is

$$\text{Final prediction} = y_1 + \hat{e}_1 + \hat{e}_2$$

x	y	y_1	e_1
70	80	100	-20
60	90	20	70
30	20	30	-10
40	40	70	-30
80	75	100	-25
90	100	90	10
10	15	25	-10

→

learning rate

↓

x	e_1	$e_1 \cdot L$ ($L = 70\%$)
70	-20	$-20 \cdot L = -14$
60	70	$70 \cdot L = 49$
30	-10	$-10 \cdot L = -7$
40	-30	$-30 \cdot L = -21$
80	-25	$-25 \cdot L =$
90	10	$10 \cdot L = 7$
10	-10	$-10 \cdot L = -7$

target for M2

x	e_2	e_2	e_3
70	-5	-7	2
60	-10	-5	-5
30	-5	-6	1
40	0	2	-2
80	-5	-8	3
90	-5	-2	-3
10	0	-1	1

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